Department of Computer Science & Engineering
The University of Texas at Arlington

Project Charter
CSE 4316: Senior Design I
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VR Wheelchair
VR Wheelchair Controller

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Revision History

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0.1	6.24.2021	EN	document creation
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1 Problem Statement

Many newly wheelchair-bound individuals face challenges in adapting to wheelchair use in real-life environments. Traditional physical therapy methods can be monotonous and may not fully prepare users for the diverse and dynamic situations they will encounter. This project aims to provide a safe, engaging, and effective way for individuals to experience and learn wheelchair use. By immersing users in a virtual environment, this solution will make the learning process more enjoyable and comprehensive, thereby enhancing their confidence and mobility skills. Solving this problem will offer a novel approach to physical therapy and improve accessibility for wheelchair users in virtual reality, leading to better preparation for real-world scenarios.

2 Methodology

The project involves the development of a system that integrates with standard long-term wheelchairs, transforming their wheels into controllers for navigating virtual reality (VR) environments. This will be achieved by designing and attaching sensor-based modules to the wheels, which capture the movement and translate it into corresponding actions within the VR system. Users will be able to spin the wheels in real life to control their movement in the virtual environment, providing an intuitive and immersive experience. The VR environments will be carefully designed to offer various scenarios, enabling users to practice and enhance their navigation skills in a safe and controlled setting.

3 Value Proposition

Investing in the VR wheelchair project offers substantial benefits to stakeholders, encompassing potential sponsors, instructors, and the university. This innovative system provides a modern, engaging approach to physical therapy, enhancing traditional methods with virtual reality technology. Here is why this project is a valuable investment:

- Enhanced Rehabilitation Outcomes: Sponsors will support a solution that positively contribute to the rehabilitation for newly wheelchair-bound individuals. By offering an immersive, virtual environment for wheelchair navigation training, the project aims to improve patient outcomes, making therapy sessions more effective and engaging. This innovative approach can lead to better skill acquisition and faster adaptation to wheelchair use in real-world scenarios.

- Research and Development Advancements: The project contributes to the growing field of VR-based therapy and assistive technology. It provides valuable insights and advancements in integrating physical and virtual interactions, offering sponsors a stake in pioneering research that could influence future developments in VR and rehabilitation technologies.

- Potential for Commercialization: The system holds potential for commercialization in the healthcare market. Sponsors could gain early access to a promising product that, upon successful development and testing, might be introduced to rehabilitation centers and healthcare providers, creating new revenue streams and market opportunities.

By supporting this project, stakeholders will be contributing to a transformative solution that enhances physical therapy for wheelchair users, advances VR technology integration, and holds a opportunity in commercialization.

4 Development Milestones

- Project Charter first draft - June 28th 2024

- System Requirements Specification - July 10th 2024

- Architectural Design Specification - July 22nd 2024
- Detailed UML Specification - August 2024
- Demonstration of Wheelchair Controller Construction - September 2024
- Demonstration of Capturing Wheelchair Movement - September 2024
- Demonstration of VR Environment - September 2024
- Demonstration of Movement Mechanism - October 2024
- Final Project Demonstration - December 2024

5 Background

Adapting to wheelchair use can be challenging for individuals who have recently become wheelchair-bound. This adaptation process involves significant physical and psychological adjustments. Traditional methods of therapy for wheelchair training typically include repetitive exercises conducted in controlled environments, such as therapy rooms. While these methods offer some benefits, they fall short in preparing users for the diverse and unpredictable situations they might encounter in daily life. As a result, many new wheelchair users experience anxiety and a lack of confidence in navigating the real world.

Existing physical therapy practices focus on basic skills such as moving the wheelchair forward, turning, and avoiding obstacles. However, these exercises are usually carried out in static, clinical settings with limited variety in scenarios. This approach does not adequately prepare users for the complex and unpredictable situations they may face outside the clinical environment, such as navigating through crowded places, dealing with uneven surfaces, or responding to emergencies. Additionally, the repetitive nature of traditional therapy can be monotonous, leading to decreased engagement and motivation among patients. This lack of engagement can slow the learning process and make it more difficult for users to apply the skills learned in therapy to real-world situations.

Virtual reality (VR) technology offers a promising solution to these challenges. VR can create immersive environments that simulate real-life scenarios, making the training more realistic and engaging. This approach provides a safe space where users can practice without the risks associated with real-world situations. VR environments can be tailored to offer a variety of scenarios, allowing users to experience different situations they might encounter and adjust their training to their individual needs and progress. This personalized and adaptable training method enhances the effectiveness and engagement of therapy. Moreover, VR-based training can incorporate gamified elements, such as challenges and rewards, to make the learning process more enjoyable. These interactive features increase motivation and participation, leading to faster skill acquisition and better preparation for real-life situations. The immersive nature of VR ensures that users develop confidence and practical skills that are directly applicable to real-world environments, thereby improving their overall quality of life.

Although there is no specific customer or sponsor currently committed to this project, the potential for future partnerships is substantial. Rehabilitation centers, healthcare providers, and technology companies could find value in adopting and supporting this innovative approach. The project's alignment with the broader goals of advancing VR-based therapy and assistive technology sets the stage for potential collaborations and investments that can drive its success and commercialization.

In summary, the VR wheelchair project presents a forward-thinking approach to improving physical therapy for newly wheelchair-bound individuals. By providing an engaging, realistic, and adaptable learning environment, it addresses the limitations of traditional training methods and enhances the overall rehabilitation experience and outcomes.

6 Related Work

Fjord's Virtual Reality System:

- Type: Enthusiast Prototype
- Description: Developed by Fjord as part of an empathy project, this system allows users to navigate a real wheelchair through virtual environments. It aims to simulate the challenges faced by wheelchair users and provide a preliminary learning experience for newly disabled individuals [1].

- Downside: This prototype is primarily geared towards creating empathy rather than offering structured training. It lacks advanced features necessary for systematic rehabilitation and skill acquisition. The system does not provide detailed feedback or adaptive training scenarios, which limits its effectiveness for long-term rehabilitation [1].

Mxtreality and WSU's AWE Simulator:

- Type: Academic Research
- Description: Mxtreality and Washington State University developed the AWE (Accessible Wheelchair Experience) simulator, a VR-based training tool for power wheelchair users. It offers customizable environments and real-time feedback to help users learn wheelchair control and improve their skills [3].
- Downside: While the AWE simulator shows promise, it is still in development and not yet commercially available. Its reliance on advanced VR hardware and software could make it expensive and potentially inaccessible for some users or facilities. Additionally, being a research project, it may not have been tested extensively in diverse real-world scenarios [2]
  .

Wheelchair Simulator VR:

- Type: Commercially available
- Description: Available on Steam, this game-like simulator provides a virtual experience where users can push a wheelchair through a city and encounter everyday challenges. It aims to raise awareness and offer a basic understanding of wheelchair navigation [2].
- Downside: Wheelchair Simulator VR focuses more on entertainment and awareness rather than serious rehabilitation. It lacks comprehensive training features and personalized feedback essential for effective skill development in actual wheelchair users. Its approach may not translate well to practical real-world applications [3]
  .

VR Wheelchair Locomotion:

- Type: Enthusiast Prototype
- Description: Built within Unity's XR Interaction Toolkit, this rig allows wheelchair locomotion in VR applications. It is designed for developers to integrate wheelchair movement in their VR projects, providing an immersive experience [4].
- Downside: As an enthusiast prototype, it is primarily a tool for VR developers rather than a finished training solution. It lacks a user-friendly interface and comprehensive training modules necessary for effective rehabilitation. Its integration requires technical expertise, limiting its accessibility for end-users and therapists [4]
  .

WheelSim:

- Type: Commercially available

- Description: WheelSim offers a virtual e-wheelchair simulator designed for learning control and traffic-related rules. It can be used on both 2D screens and in VR with Oculus goggles, providing a versatile platform for basic skills training [5].
- Downside: While WheelSim is commercially available, it mainly focuses on basic control and traffic training. It does not offer advanced rehabilitation features or adaptive learning tailored to individual needs, which can limit its effectiveness for users requiring more in-depth skill development [4].

7 System Overview

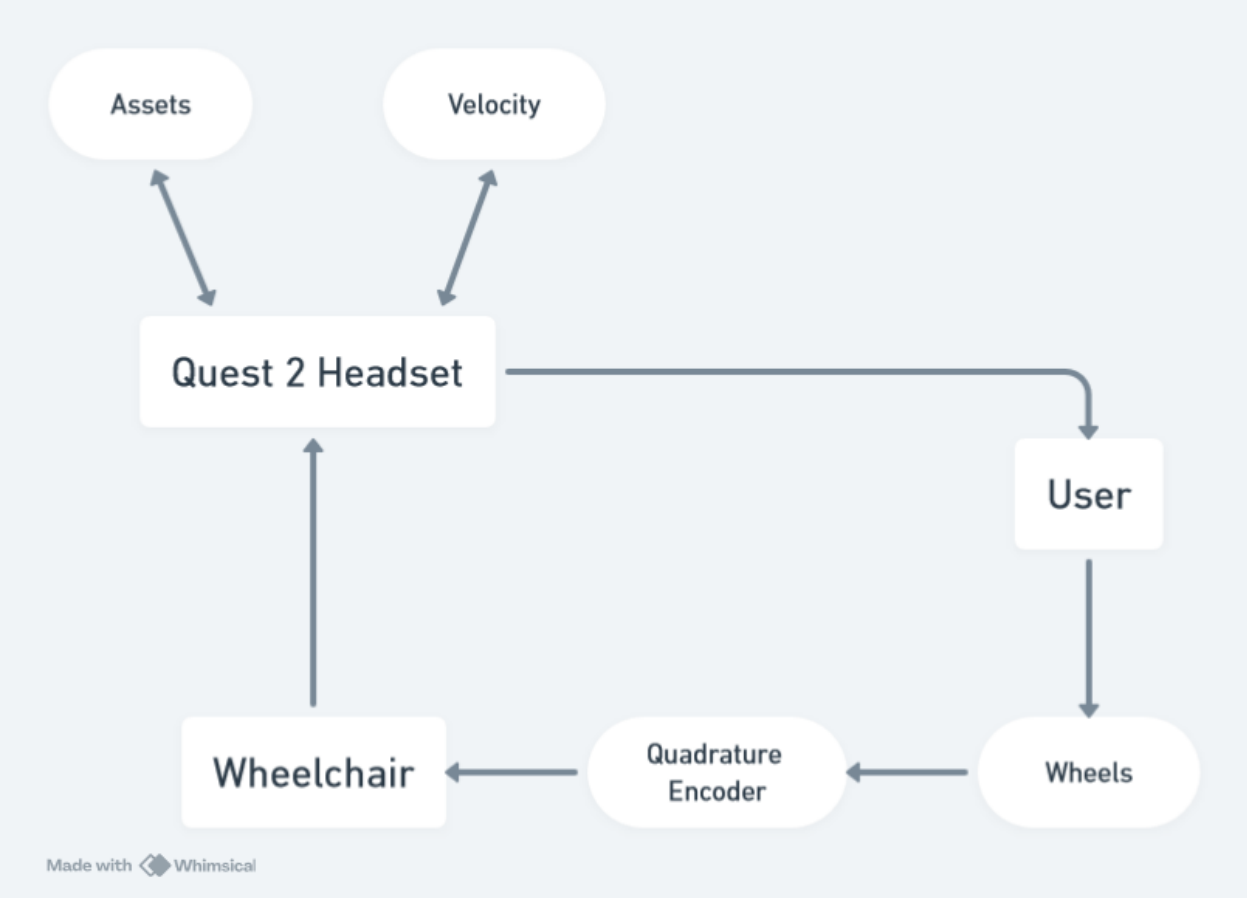


Figure 1: High-level overview of system

Our design consists of a Meta Quest 2 headset utilizing an actual, full-sized wheelchair as a controller. The wheelchair will be slightly elevated so that it sits just off the ground. This will allow for free movement of the wheels, which the user will spin to navigate in the VR environment. Quadrature encoders on each wheel will track wheel direction and velocity. This data will then be transmitted to the headset, where it will be translated into movement speed in the VR environment.

The user will sit in the wheelchair and utilize the wheels as an actual wheelchair user would to navigate various VR environments. These environments will simulate challenging situations that a wheelchair user may face in the real world. The user is forced to use only the wheels of the wheelchair to explore

these challenges and navigate an a way that is true to life. The headset will perform calculations to determine wheel velocity and travel direction, which it will then translate into accurate movement in the virtual environment.

8 Roles & Responsibilities

Title of Scrum Master is, at this point, arbitrary, and is subject to change as needed. As of now, our current stakeholders are as follows:

Shawn Gieser - Product Owner

- Point of contact and director of project

Eric Nguyen - Scrum Master, Hardware Developer

- Organize and conduct team meetings
- Design and construct hardware for wheelchair controller
- Code microcontrollers for data collection from wheelchair

Jake Cooper - Hardware Developer

- Design and construct hardware for wheelchair controller
- Code microcontrollers for data collection from wheelchair

Santiago Hernandez - Software Developer

- Create assets and environments for VR simulation
- Collect and interpret data from wheelchair controller
- Calibrate user movement within VR simulation based on user input through wheelchair

Khoa Nguyen - Software Developer

- Create assets and environments for VR simulation
- Collect and interpret data from wheelchair controller
- Calibrate user movement within VR simulation based on user input through wheelchair

9 Cost Proposal

While there is a significant amount of hardware involved in this product, costs should remain relatively low due to the availability of the Meta Quest 2 system which is provided by the department. Apart from this, the largest expense will be for the actual wheelchair, which is necessary for user experience and the fidelity of the simulation. The cost of the wheelchair is largely unknown at the moment, as research is still being done into which model or features would provide the user with the most true-to-life experience. Until a specific model is chosen, estimated cost will approximate a high average of available models.

All other substantial costs will go toward turning the wheelchair into an effective controller for the Quest 2 headset. This will require quadrature encoders and microcontrollers for each wheel, as well as various electronic components (resistors, capacitors, transistors, etc) needed to construct a working circuit board. In addition, a safe and stable platform must be constructed to elevate the wheelchair so that the wheels may spin freely.

9.1 Preliminary Budget

The following table gives a high-level overview of the anticipated product costs. Estimated costs are subject to change as necessary.

Estimated Product Costs		
Item	**Qty**	**Cost**
Wheelchair	1	500
Microcontroller	2	50
Quadrature Encoder	2	50
Circuit Board Components	Unknown	50
Platform Materials	Unknown	50
Assets/Software	Unknown	100
Total		**800**

Table 1: Overview of estimated product cost

9.2 Current & Pending Support

At this moment, the only funding available for the product is the base funding for the class, $800. Due to the potentially high cost of the wheelchair, Dr. Gieser has stated that it may be possible to procure additional funding (if needed) from other teams who do not use their full funding amount.

10 Facilities & Equipment

ERB 202, 203, 208, 335 are designated Senior Design Labs provided by the Computer Science and Engineering department. This is where we will be storage and testing our product. Furthermore, we will be using the provided 3D printer to create some custom parts that be used in the implementation of our product onto the wheelchair. Team members do have the flexibility of working on some of the task at home, but will be needing to have a computer that has access to the internet and required software.

We will be needing to purchase a micro-controller along side a quadrature encoder to record our input data of the wheels. We will also have to purchase a wheelchair to implement the hardware if one is not provided by the university.Furthermore, if necessary we will be needing other sensors or actuators if we determine a need for more data from the wheels. Software wise, we have the option of purchasing or creating new assets to put in our VR environments to simulate real world scenarios. Additionally, we may need to have software that allows the translation of the raw input date of from the sensors and micro-controller to the character in the environment.

We will borrow one the meta quest headsets that the department has in inventory. Additionally we will be borrowing the necessary equipment to develop a circuit board. Such as, the soldering iron and solder, wires, board.

11 Assumptions

The following list contains critical assumptions related to the implementation and testing of the project.

- Storage: A suitable indoor site for storage and testing
- Accuracy: The controller should be able to reliably and accurately interpret user inputs and translate them into appropriate into the environment
- Durability: The controller should be able to withstand typical usage conditions, including exposure to moisture, dust, vibrations, and temperature variations.

- Adaptability: The controller should be able to adapt to different wheelchair models and configuration. We also assumes seamless integration with other possible future features such as seating adjustments, tilt-in-space functions.
- Reliability: The Meta Quest headset and other developed hardware will continue to be functional.

12 Constraints

The following list contains key constraints related to the implementation and testing of the project.

- Hardware needs to compatible with the specified wheelchair model
- Power demand of hardware cannot be higher than supplied by standard wall outlets
- Testing must be done with a meta quest headset
- Final project demonstration must be completed by December 2024
- Total development costs must not exceed $800

13 Risks

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	Probability	Loss (days)	Exposure (days)
Loss of a team member	0.10	100	10
Illness, injuries, family emergencies	0.40	20	8
Hardware and software malfunctions and issues	0.8	80	6.4
Delays in shipping from overseas vendors	0.10	20	2.0
Testing/storage space is not available	0.05	10	.5

Table 2: Overview of highest exposure project risks

14 Documentation & Reporting

14.1 Major Documentation Deliverables

14.1.1 Project Charter

The Project Charter will be maintained and updated through Overleaf, allowing all team members to update the LaTeX document simultaneously. The initial version will be delivered on 06/26/2024 for team review, with the final version placed in the team's Google Drive and delivered on 06/28/2024 through canvas.

14.1.2 System Requirements Specification

The System Requirements Specification will be maintained and updated through Overleaf, allowing all team members to update the LaTeX document simultaneously. The initial version will be delivered on 06/26/2024 for team review, with the final version placed in the team's Google Drive and delivered on 06/28/2024 through canvas.

14.1.3 Architectural Design Specification

The Architectural Design Specification will be maintained and updated through Overleaf, allowing all team members to update the LaTeX document simultaneously. The initial version will be delivered on 07/22/2024 for team review, with the final version placed in the team's Google Drive and delivered on 07/24/2024 through canvas.

14.1.4 Detailed Design Specification

The Detailed Design Specifications will be maintained and updated through Overleaf, allowing all team members to update the LaTeX document simultaneously. The initial version will be delivered on 09/23/2024 for team review, with the final version placed in the team's Google Drive and delivered on 09/25/2024 through canvas.

14.2 Recurring Sprint Items

14.2.1 Product Backlog

The team will utilize Jira to maintain and share the backlog with its members and stakeholders. Items will be added to Jira from the SRS and be prioritized based on the next Sprint Goal. This will be decided through a group vote taking place during the sprint planning phase.

14.2.2 Sprint Planning

Each Sprint will be planned during a work day. The team will decide on the next Sprint goal based off the projects progress. Members will decide on which product backlog Items they will work on before each of the 8 sprints.

14.2.3 Sprint Goal

The group will pitch ideas and then the scrum master will hold a group vote to decide the sprint goal.

14.2.4 Sprint Backlog

The team will utilize Jira to create a scrum board of for the backlog items. During the planning the team will decide which backlog items they wish to work on, and the scrum master will be the one to finalize the items into the sprint backlog in Jira prior to the sprint.

14.2.5 Task Breakdown

Team members will start by volunteering for tasks they want to work on, and the Scrum Master will assign any remaining tasks. Each member will be responsible for documenting the time spent on their respective tasks in a spreadsheet that lists all sprint backlog items and the hours spent on each.

14.2.6 Sprint Burn Down Charts

Every team member has access to the spreadsheets through the groups Google Drive. The scrum master will assign a member to generate the burn down chart for each sprint to allow each member practice generating them.

14.2.7 Sprint Retrospective

The team will convene through zoom or in class Monday following the end of Each sprint beginning with the fifth sprint to handle the sprint retrospective. This time will be used to reflect on the sprint and finish the retrospective due the Wednesday after the end of each sprint. The team will document the sprint backlog in the google drive spreadsheet.

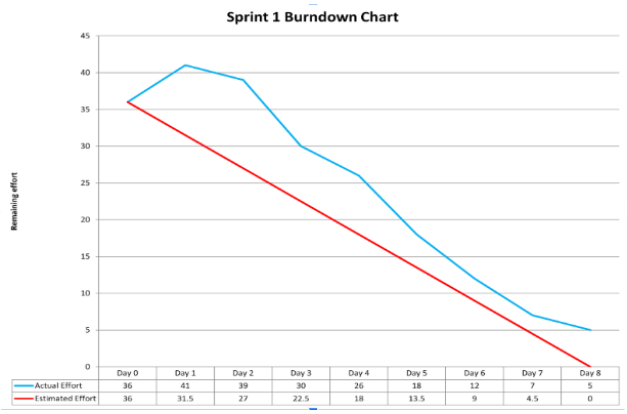


Figure 2: Example sprint burn down chart

14.2.8 Individual Status Reports

During the Friday team meeting following a sprint, each member will report all they were able to accomplish from the sprint backlog and any issues that appeared during the sprint. Along with the Sprint goal and Sprint backlog, Each individual will be required to document their individual time expenditures for their assigned tasks, a personal retrospective, and a peer review for their Individual Report.

14.3 Closeout Materials

14.3.1 System Prototype

The final system prototype will consists of a wheelchair made into a controller, a meta quest 2 headset, and a platform for which the chair will rest at an elevation. The customer will sit on the chair, wear the headset, and use the wheels of the chair to traverse various virtual environments.

14.3.2 Project Poster

Our poster, which will have final dimensions of 35.5" x 47", will include a dynamic image of a wheelchair moving along a surface that is typically not accessible. Additionally, the poster will showcase a view of one of our environments. The completed poster is scheduled to be delivered by November.

14.3.3 Web Page

Our web page will be accessible to the public and include videos of traversing our environments with its real world view as well. There will also be pictures demonstrating the projects progress. The layout of the website will be updated throughout the project but most videos will not be acquired till near closeout.

14.3.4 Demo Video

The demo videos will showcase the different terrains our wheelchair can traverse, emphasizing the ease of accessibility in these environments. The videos will include intercuts with B-reel footage of the wheelchair user navigating these terrains, highlighting how the wheels enable smooth and efficient movement. Each video will be approximately 30 seconds long, focusing on demonstrating the seamless accessibility provided by our technology.

14.3.5 Source Code

Our source code will be maintained using GitHub, which will facilitate version control and collaboration. The source code will be provided to the customer upon delivery, ensuring they have access to all necessary files for future modifications and maintenance. The project will be open-sourced under the MIT license, making it freely available to the general public. The license terms will be listed in a single README file within the repository for easy reference.

14.3.6 Source Code Documentation

The team will be using Doxygen as it supports C and C++ and the final documentation will be provided as a PDF.

14.3.7 Hardware Schematics

The project will require manually constructed circuit boards utilizing microcontrollers, quadrature encoders, and various other electronics components. However, the schematics for these boards have not yet been established.

14.3.8 CAD files

If any mechanical components are required for the wheelchair, blender will be used to generate STL files that will be provided in the teams closeout materials.

14.3.9 Installation Scripts

The team will provide installation scripts for the VR environments, and the code for the wheelchair will come pre-installed in the embedded microcontrollers.

14.3.10 User Manual

A set-up Video will be provided to the customer along with a digital user manual.

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[5] Pretorian Technologies. Wheelsim - wheelchair simulator.